Introduction to Category Theory

Eduardo Magalhães

Categories

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A *category* $\mathcal{C}$ consists of the following data:

- A collection $\mathrm{Ob}(\mathcal{C})$ of *objects*;
- For each pair of objects A, B, a collection $\hom_{\mathcal{C}}(A, B)$ whose elements are called *morphisms* or *arrows*;

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$\hom_{\mathcal{C}}(A, B)$, $\hom(A, B)$ e $\mathcal{C}(A, B)$

Examples

- **Set**, **Grp**, **Mon**, **Ring**, $\mathbf{Vect}_{\mathbb{R}}$, **Top**
- **N**
- Preorder / Partial order $(X, \leq)$
- $\mathbf{Mat}(\mathbb{R})$
- Monoid M / group G

Types of Morphisms

Definition

Let $\mathcal{C}$ be a category. A morphism $f : A \to B$ is said to be an *isomorphism* or *invertible*, if there exists a morphism $g : B \to A$ such that $gf = 1_A$ e $fg = 1_B$.
If that is the case, we say that the objects A and B are *isomorphic*, which we denote by $A \cong B$.

Note: if f is an isomorphism, then its inverse is unique, so we use the notation f^{-1}.

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Definition

A group G is a category with only one object where every morphism is an isomorphism.

Note: A category where every morphism is an isomorphism is called a *groupoid*.

Monomorphisms

Proposition

Let X, Y be sets and $f : X \to Y$ a function. The following are equivalent:

- *f is injective;*
- *Given functions g, h:*

$$\begin{tikzcd} Z \arrow[r, shift left, "g"] \arrow[r, shift right, "h"'] & X \arrow[r, "f"] & Y \end{tikzcd}$$

  If $fg = fh$, then $g = h$;
- *There exists a function $w : Y \to X$ such that $wf = 1_X$.*

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Definition

Let $\mathcal{C}$ be a category. A morphism $f : A \to B$ is a *monomorphism* or *mono* if, for any morphisms $g, h : C \rightrightarrows A$, the equality $fg = fh$ implies that $g = h$.

Epimorphisms

Proposition

Let X, Y be sets and $f : X \to Y$ a function. The following are equivalent:

- *f is surjective;*
- *Given two functions g, h:*

$$X \xrightarrow{f} Y \underset{h}{\overset{g}{\rightrightarrows}} Z$$

  If $gf = hf$, then $g = h$;
- *There exists a function $w : Y \to X$ such that $fw = 1_Y$.*

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Definition

Let $\mathcal{C}$ be a category. A morphism $f : A \to B$ is an *epimorphism* or *epic* if, for any morphisms $g, h : B \rightrightarrows C$, $gf = hf$ implies $g = h$.

Iso, Mono e Epi

Proposition

Let $\mathcal{C}$ be a category. If f is an isomorphism, then f is both a monomorphism and an epimorphism.

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Let $\mathcal{C}$ be a category. If f is an isomorphism, then f is both a monomorphism and an epimorphism.

Note: f being a monomorphism + epimorphism does not imply that f is an isomorphism.
A category where this happens is called *balanced*.

Terminal and initial objects

Definition

Let $\mathcal{C}$ be a category. An object A is called:

- *terminal* if for every object B there exists exactly one morphism $B \to A$;
- *initial* if for every object B there exists exactly one morphism $A \to B$.

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Proposition

Let $\mathcal{C}$ be a category. Then:

- *If $\mathcal{C}$ has a terminal object, then it is unique up to isomorphism;*
- *If $\mathcal{C}$ has an initial object, then it is unique up to isomorphism;*

Note: the terminal object is usually denoted by 1 and the initial object by 0 or $\emptyset$.

Opposite Category

Definition

Let $\mathcal{C}$ be a category. The *opposite category* of $\mathcal{C}$, denoted by $\mathcal{C}^{\mathrm{op}}$, is given by:

- The objects of $\mathcal{C}^{\mathrm{op}}$ are the objects of $\mathcal{C}$;
- The morphisms $\mathcal{C}^{\mathrm{op}}$ are the same as the morphisms in $\mathcal{C}$, but we 'flip' the direction. Formally:

$$\hom_{\mathcal{C}^{\mathrm{op}}}(A, B) := \hom_{\mathcal{C}}(B, A)$$

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Proposition

Let $\mathcal{C}$ be a category. Then:

- *An object A is terminal [initial] em $\mathcal{C}$ iff A is initial [terminal] in $\mathcal{C}^{\mathrm{op}}$;*
- *A morphism f is mono [epic] in $\mathcal{C}$ iff f is epic [mono] in $\mathcal{C}^{\mathrm{op}}$.*

Product of Categories

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. We define $\mathcal{C} \times \mathcal{D}$ as being the category given by:

- The objects are pairs (A, B), where $A \in \mathcal{C}$ e $B \in \mathcal{D}$. In other words $\mathrm{Ob}(\mathcal{C} \times \mathcal{D}) = \mathrm{Ob}(\mathcal{C}) \times \mathrm{Ob}(\mathcal{D})$;
- A morphism $(A, B) \to (A', B')$ is given by a pair (f, g) where $f : A \to A'$ is a morphism in $\mathcal{C}$ and $g : B \to B'$ is a morphism in $\mathcal{D}$;

Composition of two morphisms:

$$(A, B) \xrightarrow{(f,g)} (A', B') \xrightarrow{(f',g')} (A'', B'')$$

is defined as $(f', g') \circ (f, g) = (f' \circ f, g' \circ g)$.

Functors

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A *functor* $F : \mathcal{C} \to \mathcal{D}$ is given by:

- For each object $A \in \mathcal{C}$, an object $F(A)$ in $\mathcal{D}$;
- Given a morphism $f : A \to B$ em $\mathcal{C}$, a morphism $F(f)$ in $\mathcal{D}$:

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Such that $F(1_A) = 1_{F(A)}$ for every object $A \in \mathcal{C}$, and $F(f \circ g) = F(f) \circ F(g)$ for all composable morphisms $A \xrightarrow{g} B \xrightarrow{f} C$ in $\mathcal{C}$.

Functors II

Definition

Let $\mathcal{C} \xrightarrow{F} \mathcal{D} \xrightarrow{G} \mathcal{K}$ be functors. We define the composition $G \circ F : \mathcal{C} \to \mathcal{K}$ to be the functor given by:

- For any $A \in \mathcal{C}$, then $(G \circ F)(A) = G(F(A))$;
- For any morphism f in $\mathcal{C}$, then $(G \circ F)(f) = G(F(f))$.

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Definition

We define **Cat** as being the category whose objects are categories, and morphisms are functors, where composition is defined as above.

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Definition

Two categories $\mathcal{C}$ and $\mathcal{D}$ are said to be isomorphic if they are isomorphic as objects in **Cat**. In other words, if there are functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ such that $GF = 1_{\mathcal{C}}$ e $FG = 1_{\mathcal{D}}$.

Full and Faithful

Note: Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. For each pair of objects $A, B \in \mathcal{C}$, the functor F induces a map

$$\begin{aligned} F : \hom_{\mathcal{C}}(A, B) &\to \hom_{\mathcal{D}}(F(A), F(B)) \\ f &\mapsto F(f) \end{aligned} \tag{1}$$

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A functor $F : \mathcal{C} \to \mathcal{D}$ is said to be:

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Proposition

Let $F : \mathcal{C} \to D$ be a functor. Then F is an isomorphism iff it is bijective on objects, and fully faithful.

Functors III

Proposition

Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. Then:

1. *F preserves isomorphisms;*
2. *If F is fully faithful, then F reflects isomorphisms;*
3. *If F is faithful, then F reflects monomorphisms and epimorphisms.*

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Corollary

Let $F : \mathcal{C} \to \mathcal{D}$ be a functor and A, B objects in $\mathcal{C}$. If $A \cong B$, then $F(A) \cong F(B)$, where the converse is true if, for example, F is fully faithful.

Contravariant Functors

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A *contravariant functor* $F : \mathcal{C} \to \mathcal{D}$ is given by:

- For each object $A \in \mathcal{C}$, an object $F(A)$ in $\mathcal{D}$;
- Given a morphism $f : A \to B$ em $\mathcal{C}$, a morphism $F(f)$ in $\mathcal{D}$:

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A contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ is the same thing as a functor $F : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$.

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Proposition

The assignment $\mathcal{C} \mapsto \mathcal{C}^{\mathrm{op}}$ and $F \mapsto F$ defines a functor **Cat** $\to$ **Cat**.